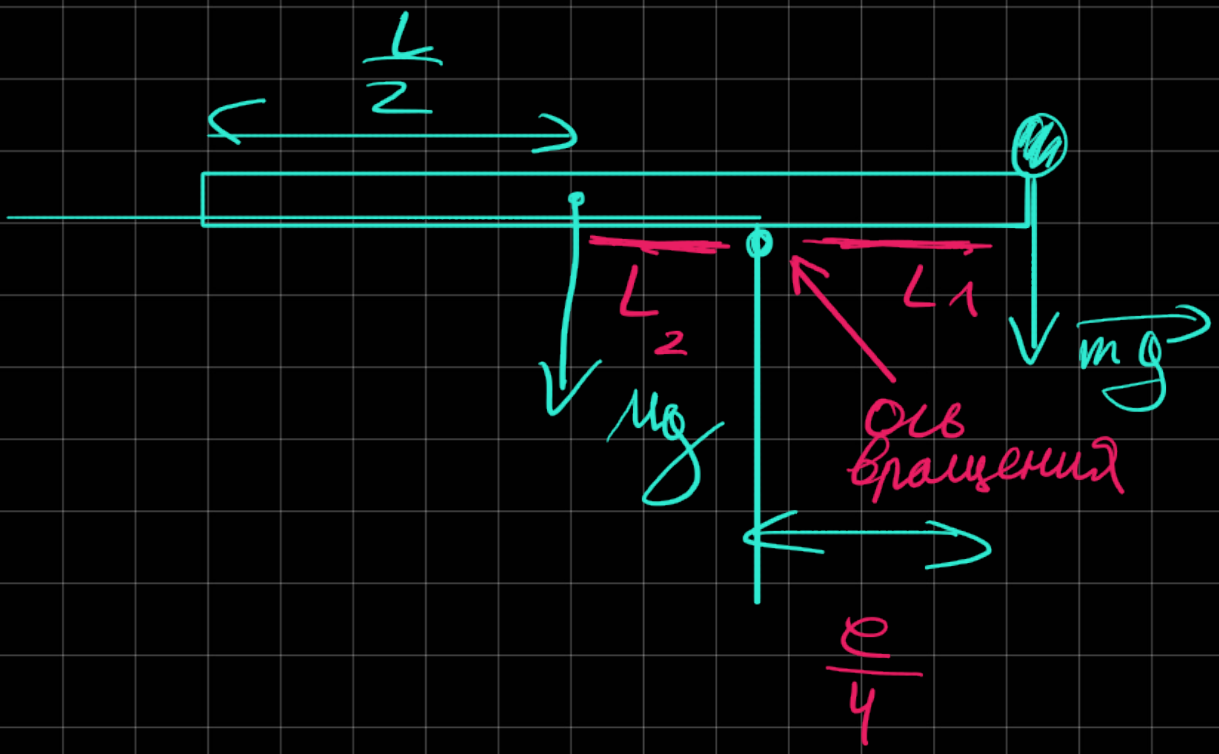
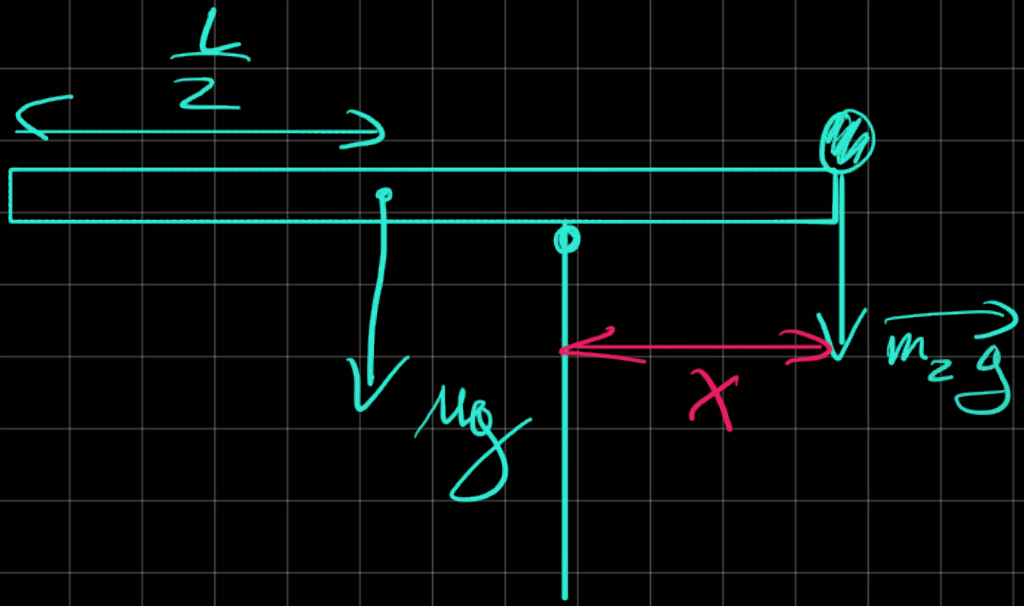
$L = 60\,см = 0{,}6\,м \qquad m_1 = 250_2 = 0{,}25_2$



$$\Sigma \vec{M} = I\vec{\varepsilon} = 0$$

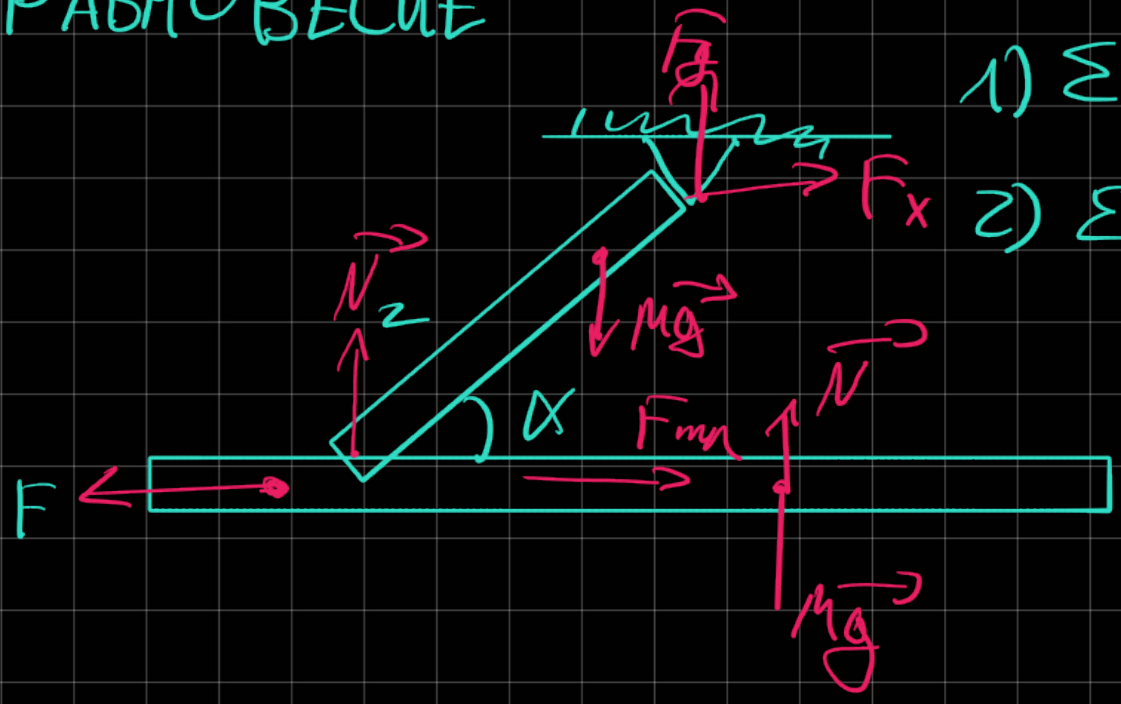


$$M_{m_2 g} = M_{mg}$$

$$m_2 g x = Mg\left(\frac{L}{2} - x\right)$$

$$m_2 x = \frac{Me}{2} - Mx$$

РАВНОВЕСИЕ



1) $\Sigma \vec{F} = 0$

2) $\Sigma \vec{M} = 0$

1) $\vec{F} + \vec{F}_{тр} + \vec{N} + \vec{Mg} + \vec{P} = 0$

т.к $V = const \Rightarrow a = 0$

X: $F - F_{тр} = 0 \Rightarrow F = F_{тр}$

O: $M_{mg} = M_N + M_{F_{тр}}$

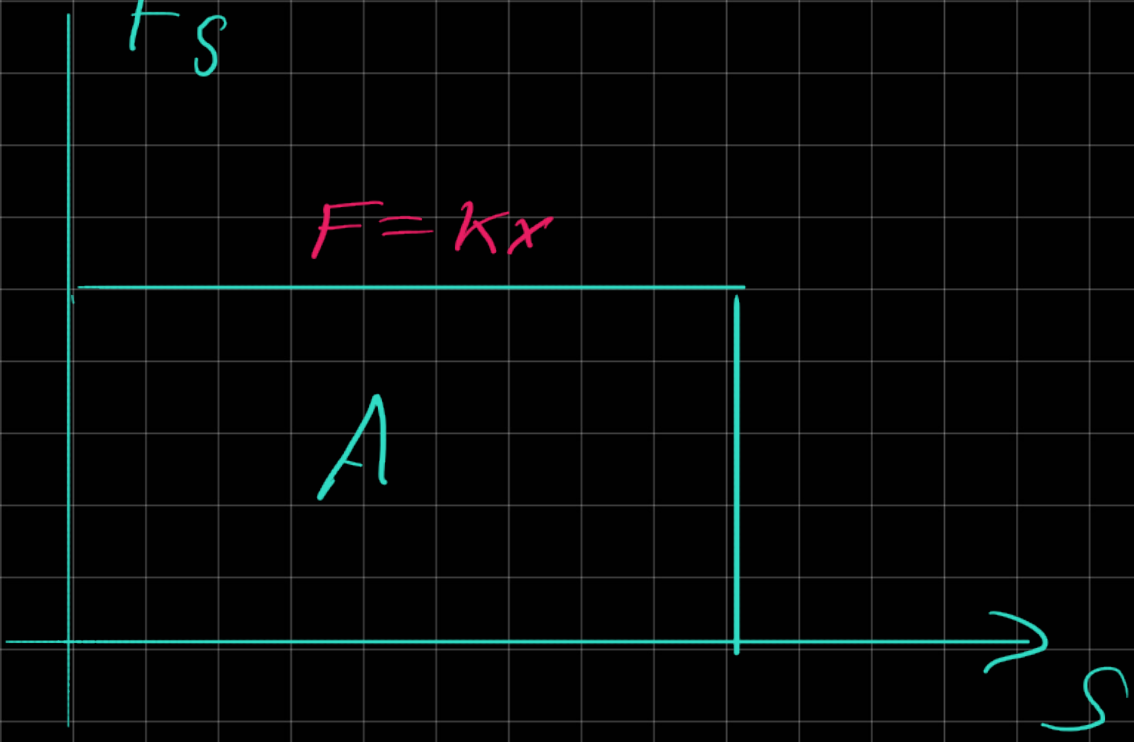
$\cos\alpha = \frac{L_1 : 2}{L} \Rightarrow L_1 = \frac{L}{2} \cdot \cos\alpha$

$L_1 = \frac{L}{2} \cdot \frac{1}{2}$

РАБОТА

$$A = \vec{F} \cdot \vec{S}$$

$$[A] = Н \cdot м = Дж$$





$$A = \frac{kx \cdot x}{2}$$

Энергия

Кинетическая

$$E_к = \frac{m \cdot v^2}{2}$$

Потенциальная

$$E_п = mgh$$

$$E_п = \frac{kx^2}{2}$$

$E_к + E_п = E$ } полная механическая энергия тела

$$A = \Delta E_к$$

$$A = -\Delta E_п$$

